

L5 Nested Loops

For Discord email - support@learnyard.com

if (condition)

{

if (condition)

{

}

}

Nested if - else

for (int i = 1; i <= n; i++)

{

for (int j = 1; j <= n; j++)

{

}

}

Nested
loop

```
for (int i = 1; i <= 3; i = i + 1)
```

```
{
```

```
for (int j = 1; j <= 3; j = j + 1)
```

```
{
```

```
    S.O.Pln ("Hello");
```

```
}
```

```
}
```

$i = 1, j \rightarrow 3$

$i = 2, j \rightarrow 3$

$i = 3, j \rightarrow 3$

3 times
print

$i = 1, 1 <= 3$ (true)

$j = 1; j <= 3, \text{Hello}$

$j = 2; j <= 3; \text{Hello}$

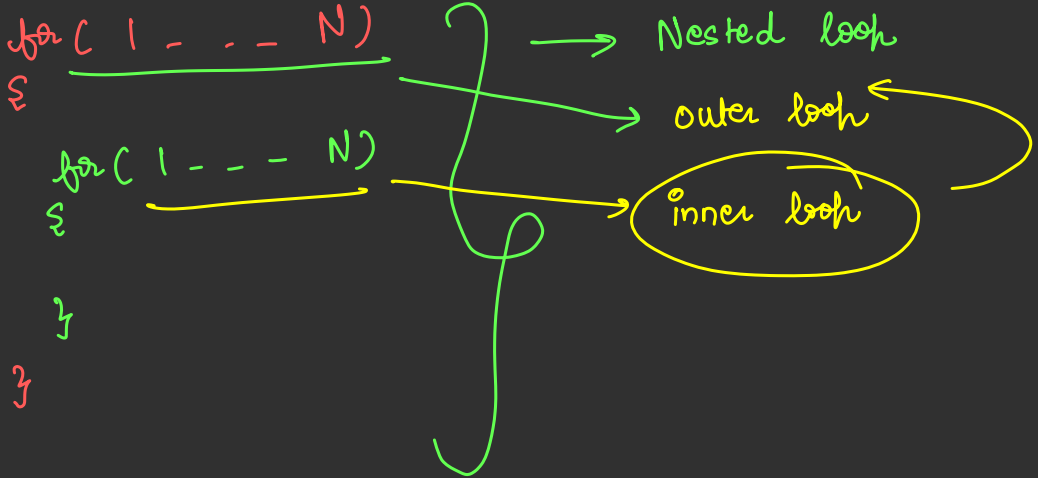
$j = 3; j <= 3; \text{Hello}$

$i = 2, 2 <= 3$ (true)

$j \rightarrow 1 \dots 3$

$i = 3, 3 <= 3$ (true)

What are Nested Loops ?



Dry run of basic Nested Loop

for (int i = 1; i <= 3; i = i + 1)

{

for (int j = 1; j <= 3; j = j + 1)

{

S.D.Pln(j);

}

}

1
2
3
1
2
3

1
2
3

j = 4

4 <= 3

↓
false

i = 1, j → 1, 2, 3

i = 2, j → 1, 2, 3

i = 3, j → 1, 2, 3

for (
 ε

) → Row change

Number
of lines

for (

)

→

What it is
printed in
row?

↓
enter loop

}

line |
column

rows

* → 1 star
* * → 2 stars
* * * → 3 stars
* * * *
↓
4 stars

for (int i = 1; i <= 3; i = i + 1)
{
 ↳ lines →

for (int j = 1; j <= 2; j = j + 1)

{
 S.O.P (" * "); → No change
 in line
}

S.O.Pln (); → line change
}

j = 1, 2 → 2 star

i = 1 ↘

* *

i = 2 ↘

* *

i = 3

* *

Square Pattern

$N = 5$

N row

```
for (int i = 1; i <= N; i = i + 1)
```

{

```
for (int j = 1; j <= N; j = j + 1)
```

{

```
    S.O.P ("*");
```

}

```
    S.O.Pln();
```

}

Every line

will have n star

Input

Input number of rows: 5

Output

```
*****  
*****  
*****  
*****  
*****
```

row = N

column = N

$N \rightarrow \text{row} \rightarrow \text{outer loop}$

$\text{star} = 1, \text{space} = 3$

```
for (int i = 1; i <= 4; i = i + 1)
```

```
{ for (int j = 1; j <= space; j = j + 1)
```

```
{ S.O.P (" ");
```

```
for (int j = 1; j <=  $\text{star}$ ; j = j + 1)
```

```
{ S.O.P (" * ");
```

```
}
```

```
S.O.Pln ();
```

$\text{star} = \text{star} + 1;$ $\text{space} = \text{space} - 1;$

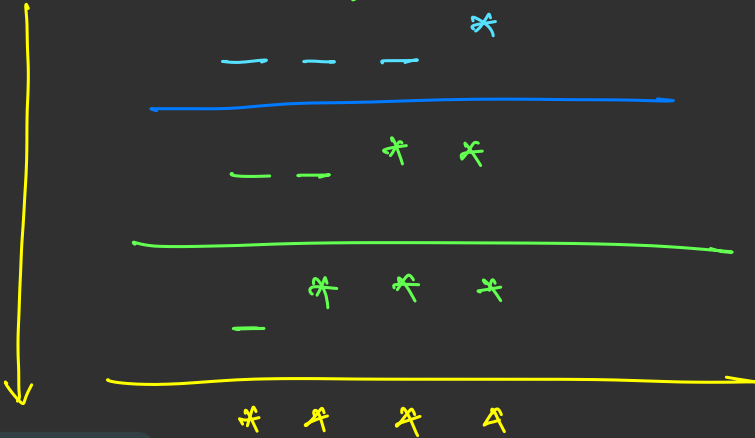
Empty space

—	—	—	*	→ 1
—	—	*	*	→ 2
—	*	*	*	→ 3
*	*	*	*	→ 4

$i = 1$, $star = 1$, $space = 3$
~~2~~ 4 ~~2~~ 0
~~3~~

space ↓

star ↑



Rhombus Pattern

space = 0
N lines

Outer loop → row

for (int i = 1; i <= n; i++)
{

// space → 1 --- space

// star → for (int j = 1; j <= N; j++)
{
 S.O.P (" ");
}

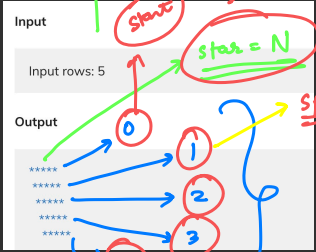
S.O.Pln ();

Space = space + 1;

}

for (int j = 1; j <=
space; j++)

S.O.P (" ");



N = 5

start

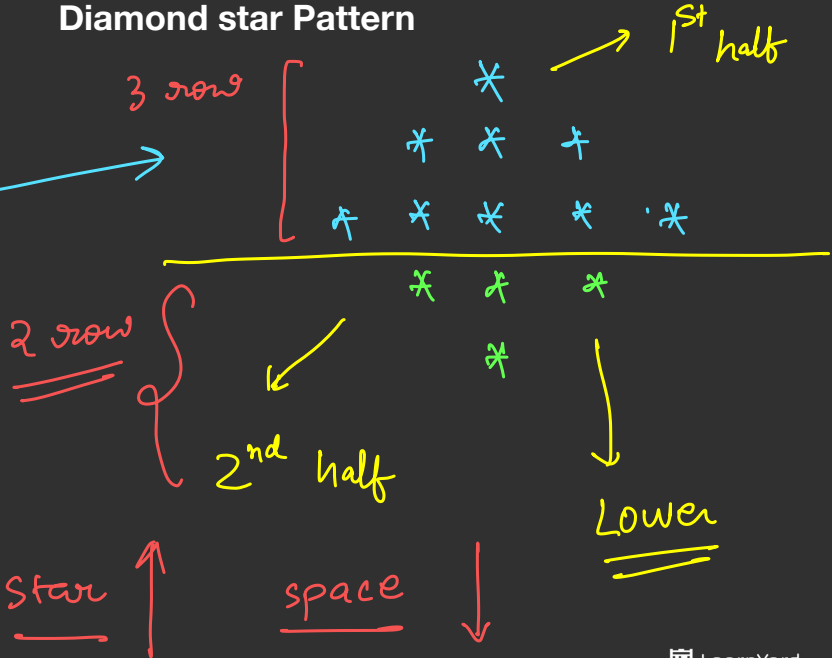
star = N

space

Diamond star Pattern

```
Enter N: 5
  *
 ***
*****
*****
*****
  ***
   *
```

Handwritten annotations on the code block:
- A bracket on the right side of the first 5 rows is labeled "5".
- A bracket on the right side of the last 3 rows is labeled "3".



N = 3

4 → _ _ _ _ * → 4 space, 1 star
 2 → _ _ * _ _ * _ * → 2 space, 3 star
 0 → * _ * _ * _ * _ * → 0 space, 5 star
spaces

space = space - 2
 star = star + 2

Upper

Discussion on Number Patterns

1	2
12	3
123	4
1234	5
12345	

1
21
321
4321
54321

Input
Input N: 5
Output
5
44
333
2222
11111

N → row N = 3

for (int i = 1; i <= N; i = i + 1)

{

for (int j = 1; j <= i; j = j + 1)

{

S.O.P(j);

}

S.O.Pln();

}

i = 1, j → 1 to 1

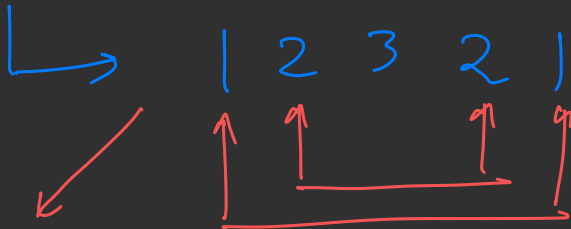
i = 2, j → 1 to 2

i = 3, j → 1 to 3

reverse
↑
number

While Loop Question

Check whether a number is palindrome or not



Palindrome

original

1 2 3

reverse

3 2 1

Practice and Homework Questions

Homework questions -

<https://codeforwin.org/c-programming/c-program-to-print-number-pattern-with-1-0-at-alternate-columns>

<https://codeforwin.org/c-programming/number-pattern-22-in-c>

<https://codeforwin.org/c-programming/equilateral-triangle-star-pattern-program-in-c>

<https://codeforwin.org/c-programming/c-program-to-print-x-star-pattern>

Thank You!

Please practice more questions and examples as above !!